

RESEARCH PROPOSAL

Behavior-Selected Eigenspace Compression for Efficient Answer Readout in Continuous-Reasoning Language Models

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Abstract

Continuous-reasoning language models replace long written chains of thought with a small number of continuous hidden states. This reduces the number of generated reasoning tokens, but the model must still convert its final 768-dimensional state into scores for all 50,257 vocabulary tokens. This proposal investigates whether the answer-bearing part of that state can be identified through covariance-matrix eigendecomposition and used to construct a lower-cost vocabulary readout. Preliminary experiments on the released CODI GPT-2 checkpoint identified 28 eigendirections, indexed 4 through 31, at the final answer-cue state. Although these directions explain only 11.31 percent of state variance, retaining them preserves 87.8 percent of baseline numeric exact-match accuracy, while removing them reduces accuracy by 30.48 percentage points. A rank-32 variant preserves 94.4 percent of baseline accuracy. By projecting the vocabulary matrix into the selected eigenspace offline, the online readout can operate on 28 or 32 coordinates rather than 768. An isolated Tesla T4 benchmark measured the full head at 1,327 microseconds and the rank-32 head at 117 microseconds, an 11.3-fold component-level speedup. The proposed research will integrate this readout into end-to-end generation, compare it against established compression methods, and test whether behavior-selected eigendirections transfer across tasks and checkpoints.

1. Introduction

Language models normally express reasoning as text. Each reasoning token requires another pass through the transformer and another vocabulary prediction. CODI, or Continuous Chain-of-Thought via Self-Distillation, instead learns a short sequence of continuous thoughts and aligns the student's internal answer representation with that of an explicit-reasoning teacher [1]. The released CODI system used in this project is based on GPT-2 small [2]. It contains twelve transformer blocks, has a representation width of 768, and uses a vocabulary of 50,257 tokens. During CODI inference, a question is encoded, six continuous latent iterations are executed through the shared transformer, and the model then receives the answer cue. The final representation at the colon in 'The answer is:' is transformed by the language-model head into one score for every vocabulary token.

The language-model head is therefore a natural optimization target. Its learned vocabulary matrix can be viewed as 50,257 token-specific detectors, each acting on the 768 coordinates of the final hidden state. A normal readout calculates a dense matrix product for every generated token. If answer choice at this endpoint depends mainly on a much smaller behaviorally important subspace, the head can be reorganized so that only coordinates in that subspace are used online.

2. Problem Statement

Existing low-rank compression commonly preserves directions with high variance or minimizes reconstruction error. Those objectives need not preserve task accuracy. Our preliminary CODI result demonstrates a strong separation between variance and answer contribution: the first four

eigendirections contain 82.31 percent of variance but preserve only 6.1 percent of baseline exact-match accuracy when retained alone. In contrast, eigendirections 4 through 31 contain only 11.31 percent of variance but preserve 87.8 percent of baseline accuracy. A compression method that blindly keeps only the highest-variance directions can therefore discard the dimensions that matter for the answer.

The research problem is to determine whether behavior-selected eigendirections can support a practical, reliable, and generalizable low-rank answer readout for continuous-reasoning models. The central challenge is not simply reducing matrix size. It is preserving the model's answer ranking while proving that any measured speedup survives integration into the complete autoregressive generation pipeline.

3. Aim, Objectives, and Research Questions

Aim

To develop and evaluate an eigenspace-guided method for reducing the computational cost of CODI's vocabulary readout while retaining mathematical reasoning accuracy.

Objectives

1. Identify stable answer-bearing eigendirections at CODI's final answer-cue representation using held-out calibration and causal retention/removal interventions.
2. Convert the selected eigenspace into rank-28 and rank-32 language-model heads through an offline transformation of the learned vocabulary matrix.
3. Measure exact-match accuracy, token-ranking agreement, numerical error, memory use, and end-to-end latency.
4. Compare behavior-selected eigendecomposition with weight-only decomposition, variance-ranked selection, activation-aware compression, and vocabulary-shortlisting baselines.
5. Test robustness across data splits, random seeds, mathematical reasoning datasets, and compatible checkpoints.

Research questions

- How many eigendirections are sufficient and necessary for CODI's answer accuracy at the forced answer cue?
- Does behavior-based eigendirection selection preserve accuracy better than selection based only on variance or matrix reconstruction error?
- Does the reduction in language-model-head operations produce a measurable end-to-end inference improvement?
- Are the selected eigendirections stable across calibration samples, tasks, and model checkpoints?

4. Background and Related Work

4.1 Continuous reasoning and CODI

CODI trains a shared model in teacher and student modes. The teacher receives an explicit chain of thought, whereas the student performs a fixed number of continuous latent iterations. Training combines language-model objectives with alignment between teacher and student representations at the final answer-generating position [1]. At inference, the teacher branch is not required: the student processes the question, performs the latent iterations, consumes the answer cue, and generates answer tokens through the normal vocabulary head.

4.2 Low-rank and activation-aware compression

Low-rank factorization replaces one large linear map with two smaller maps. Fisher-weighted factorization showed that minimizing ordinary reconstruction error can be misaligned with task accuracy and proposed weighting parameters by their effect on model predictions [4]. SliceGPT uses orthogonal transformations and post-training dimensional reduction across transformer representations [5]. ASVD adapts matrix decomposition to observed activation distributions [6], while FLAT-LLM performs training-free transformations using eigendirections of activation spaces [7]. These studies establish that model behavior and activation statistics should influence compression, but they do not test CODI’s answer-cue endpoint or the particular non-leading eigenspace identified here.

4.3 Vocabulary-head acceleration

Adaptive softmax reduces expected vocabulary computation by exploiting token-frequency structure [8], and vocabulary clustering predicts a smaller set of candidate tokens before projection [9]. SlimSpec directly applies a low-rank vocabulary head to a speculative-decoding draft model while retaining the full vocabulary [10]. ARCHead combines a compressed output head with an activation-derived metric [11]. These approaches confirm that the vocabulary projection can be a meaningful bottleneck. The proposed study differs by selecting the hidden subspace using causal answer-accuracy interventions in a continuous-reasoning model rather than only frequency, reconstruction error, or a learned auxiliary objective.

5. Preliminary Work Completed

5.1 Reproduction and state collection

The author-released zen-E/CODI-gpt2 checkpoint was reproduced in a pinned software environment. Native GSM8K numeric exact match was 0.4359. Under the forced answer cue used for intervention, baseline accuracy was 0.4337, corresponding to 572 correct answers among 1,319 test questions. All confirmatory runs used float32. For eigenspace construction, final answer-cue representations were collected from 2,048 GSM8K training questions. Each representation was the 768-dimensional state labelled state 12, produced after the twelve GPT-2 blocks at the colon in the answer cue and immediately before vocabulary readout.

5.2 Covariance eigendecomposition

Let each collected endpoint representation be h_i and let μ be their mean. The centered rows were stacked into a matrix X . The 768 by 768 covariance matrix was then constructed as:

$$C = (1 / (n - 1)) X^T X$$

Because C is symmetric, it can be decomposed into orthonormal eigendirections and their associated eigenvalues:

$$C = U \Lambda U^T$$

The columns of U define directions in the original 768-dimensional representation space. The eigenvalues in Λ measure how strongly the calibration states vary along those directions. Retention interventions replaced the unselected component of a test state with the calibration mean. Removal interventions did the reverse. Full greedy decoding was then used to measure numeric exact match, so the result reflects complete answers rather than only the first generated token.

5.3 Discovery of 28 answer-bearing directions

Condition	Dimensions	Variance share	Exact match	Baseline retained
Unmodified baseline	768	100.00%	0.4337	1.000
Retain eigenvectors 0-3	4	82.31%	0.0265	0.061
Retain eigenvectors 4-15	12	7.49%	0.2191	0.505
Retain eigenvectors 4-31	28	11.31%	0.3806	0.878
Retain eigenvectors 0-31	32	93.62%	0.4094	0.944
Remove eigenvectors 4-31	28 removed	11.31%	0.1289	0.297
Remove eigenvectors 0-3	4 removed	82.31%	0.4185	0.965

Table 1. Confirmatory full-GSM8K results for interventions at the answer-cue endpoint.

All preregistered confirmation conditions passed. Retaining eigendirections 4-31 produced 0.3806 exact-match accuracy, equal to 87.8 percent of the 0.4337 baseline. Removing the same 28 directions reduced accuracy to 0.1289, a loss of 30.48 percentage points; the paired confidence interval was +27.90 to +33.13 points and the exact McNemar p-value was 5.7×10^{-101} . Four matched random rank-28 subspaces retained only 6.5 to 8.7 percent of baseline accuracy. The selected band was also stable across disjoint calibration halves, with mean principal-angle cosines of 0.979 and 0.968.

These results establish a bounded mechanistic claim: at this frozen CODI checkpoint, dataset, endpoint, and intervention type, a 28-dimensional linear subspace is both highly sufficient and

strongly necessary for answer accuracy. The experiment does not establish that the same directions are universal, nonlinear, or independently interpretable. Retention also moves states away from their ordinary distribution because discarded coordinates are replaced by the mean.

6. Proposed Eigenspace Readout

6.1 Standard language-model head

Let h be the final 1 by 768 hidden state and W be the learned 50,257 by 768 vocabulary matrix. The standard head calculates:

$$z = h W^T$$

The result z contains 50,257 logits. Each logit is a score for one vocabulary token. Greedy decoding selects the token with the largest score. This multiplication requires $768 \times 50,257 = 38,597,376$ multiply-accumulate operations for one state.

6.2 Offline head transformation

Choose r columns of U and place them in U_r , where r is 28 or 32. Before inference, project every vocabulary detector into the selected eigenspace:

$$A = W U_r$$

For rank 28, A has shape 50,257 by 28. For rank 32, it has shape 50,257 by 32. The centering contribution is also computed once as $b = \mu W^T$. Neither A nor b depends on the incoming question, so both can be stored with the optimized model.

6.3 Online inference

For each final hidden state, inference performs two smaller operations:

$$c = (h - \mu) U_r$$

$$z_{\text{hat}} = b + c A^T$$

The first operation converts 768 coordinates into r eigenspace coordinates. The second converts those coordinates into scores for the full vocabulary. The full vocabulary remains available; the approximation restricts how hidden-state information reaches the vocabulary matrix. Rank 28 corresponds exactly to the discovered answer-bearing band 4-31. Rank 32 additionally includes directions 0-3 and achieved higher retained accuracy in the preliminary intervention.

Readout	Online MACs per token	Theoretical reduction	T4 time
Full 768-dimensional head	38,597,376	1.00x	1,327 microseconds
Rank-28 eigenspace head	1,428,700	27.0x fewer	120 microseconds
Rank-32 eigenspace head	1,632,800	23.6x fewer	117 microseconds
Rank-64 eigenspace head	3,265,600	11.8x fewer	154 microseconds

Table 2. Isolated language-model-head measurements on a Tesla T4 in float32. These are component timings, not end-to-end generation speedups.

The theoretical multiply-accumulate reduction does not translate one-for-one into elapsed-time reduction because memory movement, kernel launch cost, batching, and the transformer blocks also consume time. Nevertheless, the component benchmark shows that the factorized computation can be materially faster when implemented as dense matrix operations.

6.4 Latent-budget result

A separate preliminary efficiency experiment reduced CODI's six continuous latent iterations. On full GSM8K, the trained $M=6$ setting scored 0.4337, while $M=5$ scored 0.4359, a difference of +0.23 percentage points. $M=4$ and $M=3$ fell to 0.3836 and 0.3768. Thus one latent iteration could be removed without an observed accuracy cost, but batch-32 wall-clock time on the T4 was essentially flat because fixed costs dominated. This result will be re-evaluated under latency-bound batch-1 inference and combined with the low-rank readout only after each intervention is validated independently.

7. Proposed Methodology

7.1 Experimental design

- Freeze independent training/calibration, eigenspace-selection, validation, and final test partitions to prevent using test performance to choose directions or rank.
- Collect answer-cue state-12 representations and compute the covariance matrix and its eigendecomposition in float32.
- Select candidate direction bands using only the selection partition and preregister the final rank and intervention thresholds.
- Construct rank-28, rank-32, and rank-sweep heads offline and integrate them into every generated-answer step of the released CODI decoder.
- Evaluate correctness and efficiency on the untouched test sets. Repeat the process across seeds, compatible checkpoints, and additional mathematical reasoning datasets.

7.2 Comparison methods

- The original full 768 by 50,257 vocabulary head.
- Leading-eigenvalue rank- r selection without behavioral intervention.
- Weight-only truncated decomposition of the vocabulary matrix.
- Activation-aware and Fisher-weighted low-rank baselines.
- A learned rank- r head initialized from the offline eigenspace factors.
- Vocabulary shortlisting or clustering as a different approach to the same output bottleneck.

7.3 Outcome measures

- Primary quality outcome: numeric exact-match accuracy with paired confidence intervals.

- Readout fidelity: top-1 and top-k token agreement, logit error, cross-entropy, and answer-margin preservation.
- Efficiency: batch-1 and batch-32 latency, tokens per second, peak accelerator memory, head time, and complete generation time.
- Robustness: eigenspace overlap, retained accuracy, and latency across calibration resamples, tasks, seeds, and checkpoints.

7.4 Statistical analysis

All accuracy comparisons will be paired by question. Exact McNemar tests will measure discordant correctness outcomes, and paired bootstrap intervals will quantify accuracy differences and retention ratios. Multiple planned comparisons will use a declared correction procedure. Direction selection will not be changed after the test set is opened. Hardware timing will include warm-up, synchronization, repeated trials, medians, dispersion, and fixed software and precision settings.

8. Expected Contributions

- A mechanistic account of where CODI's final answer information is concentrated at its answer cue.
- A behavior-selected alternative to compression based only on eigenvalue magnitude or reconstruction error.
- A deployable low-rank vocabulary head that retains the complete token vocabulary.
- A careful separation between component-level speedup and real end-to-end inference improvement.
- A reproducible evaluation protocol for connecting hidden-state interventions to systems optimization.

9. Risks and Limitations

The strongest preliminary evidence is limited to one released CODI GPT-2 checkpoint, GSM8K, one semantic endpoint, and linear interventions. A subspace selected on one task can overfit task vocabulary or answer format. Mean replacement can create hidden states that are not naturally produced by the model. The rank-28 readout loses approximately 5.3 absolute accuracy points relative to baseline, while rank 32 loses approximately 2.4 points. The isolated 11.3-fold head speedup cannot be reported as an end-to-end model speedup. Furthermore, an effective rank-r output map can alter which vocabulary tokens are reachable as top-ranked outputs. These risks require rank sweeps, learned correction, answer-token coverage analysis, broader datasets, and fully integrated timing.

10. Work Plan

Phase	Work	Planned output
1	Reproduce all preliminary measurements and freeze data splits.	Audited baseline and artifact manifest
2	Implement rank-28 and rank-32 heads in the generation loop.	End-to-end inference implementation
3	Run rank sweeps and comparison methods.	Accuracy, agreement, and latency results
4	Evaluate transfer across tasks, seeds, and checkpoints.	Generalization and failure analysis
5	Write the final study and release reproducible artifacts.	Dissertation paper and code package

11. Conclusion

The preliminary study found that CODI's final 768-dimensional answer representation is not equally important in all directions. A specific set of 28 covariance eigendirections carries a small share of total variance but most of the model's answer accuracy. This finding suggests a direct systems application: project both the incoming hidden state and the learned vocabulary detectors into the behaviorally selected eigenspace, thereby replacing a large vocabulary multiplication with two smaller operations. Early component measurements are promising, but the academic contribution depends on rigorous end-to-end evaluation and comparison with established compression methods. The proposed research will determine whether this mechanistic finding can become a reliable inference optimization rather than remaining an isolated observation.

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